

Exercises from chapter 5

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Exercise 1

Suppose by absurdity that $\exists w \in \mathbb{R}^n$ such that $Aw = 0w$, that is, $\lambda = 0$ is an eigenvalue and w is the respective eigenvector with $w \neq 0$. So:

$$Aw = 0 \implies w^\top Aw = w^\top 0 = 0$$

But $w^\top Aw > 0$. Absurd. So A doesn't have a 0 eigenvalue. Therefore A is invertible

Exercise 2

We can show that the coercicity and continuity constants are 1, when using the energy norm:

Continuity constant

Using the Cauchy-Schwarz inequality:

$$|a(u, v)| \leq \|u\|_a \|v\|_a \implies N_a = 1$$

Coercity constant

We just need to note that $a(u, u) = \|u\|_a^2 = \alpha \|u\|_a^2$. So $\alpha = 1$
Applying this to Céa's lemma, we obtain:

$$\|u - u_h\|_a \leq \|u - v_h\|_a, \forall v_h \in V_h$$